

October 8, 2026

1 Analysis of the risk of failure of the O-rings on the Challenger shuttle

On January 27, 1986, the day before the takeoff of the shuttle *Challenger*, had a three-hour tele-conference was held between Morton Thiokol (the manufacturer of one of the engines) and NASA. The discussion focused on the consequences of the temperature at take-off of 31°F (just below 0°C) for the success of the flight and in particular on the performance of the O-rings used in the engines. Indeed, no test had been performed at this temperature.

The following study takes up some of the analyses carried out that night with the objective of assessing the potential influence of the temperature and pressure to which the O-rings are subjected on their probability of malfunction. Our starting point is the results of the experiments carried out by NASA engineers during the six years preceding the launch of the shuttle Challenger.

1.1 Loading the data

We start by loading this data:

```
[20]: import numpy as np
import pandas as pd
data = pd.read_csv("shuttle.csv")
# We now take into consideration all data
```

```
[20]:
```

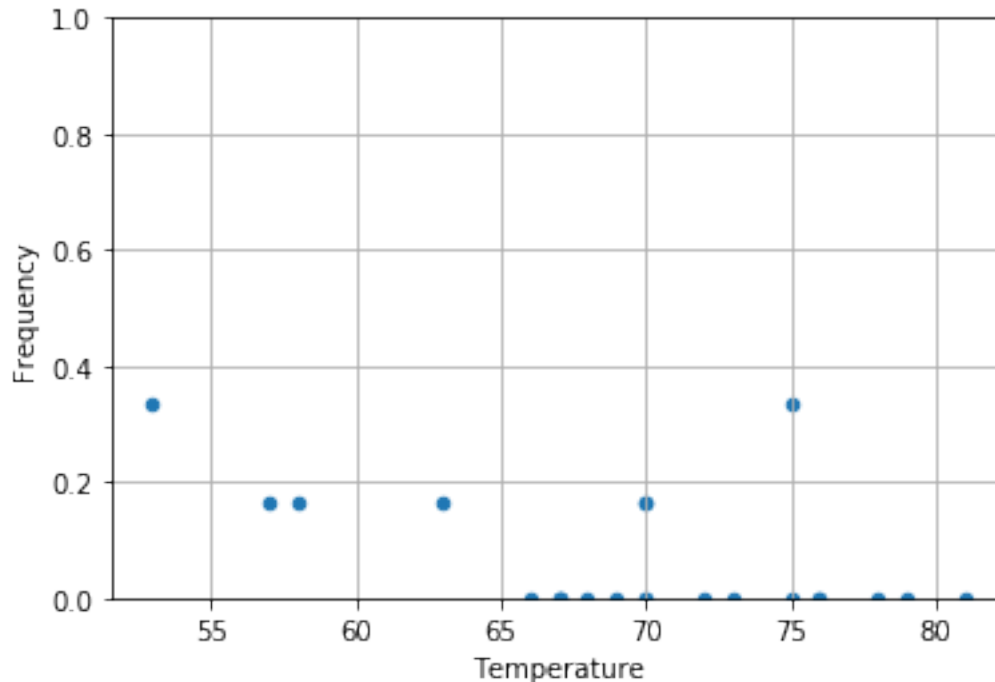
	Date	Count	Temperature	Pressure	Malfunction
0	4/12/81	6	66	50	0
1	11/12/81	6	70	50	1
2	3/22/82	6	69	50	0
3	11/11/82	6	68	50	0
4	4/04/83	6	67	50	0
5	6/18/82	6	72	50	0
6	8/30/83	6	73	100	0
7	11/28/83	6	70	100	0
8	2/03/84	6	57	200	1
9	4/06/84	6	63	200	1
10	8/30/84	6	70	200	1
11	10/05/84	6	78	200	0
12	11/08/84	6	67	200	0

13	1/24/85	6	53	200	2
14	4/12/85	6	67	200	0
15	4/29/85	6	75	200	0
16	6/17/85	6	70	200	0
17	7/29/85	6	81	200	0
18	8/27/85	6	76	200	0
19	10/03/85	6	79	200	0
20	10/30/85	6	75	200	2
21	11/26/85	6	76	200	0
22	1/12/86	6	58	200	1

The data set shows us the date of each test, the number of O-rings (there are 6 on the main launcher), the temperature (in Fahrenheit) and pressure (in psi), and finally the number of identified malfunctions.

```
[21]: %matplotlib inline
pd.set_option('mode.chained_assignment',None) # this removes a useless warning
↳ from pandas
import matplotlib.pyplot as plt

data["Frequency"] = data.Malfunction/data.Count
data.plot(x="Temperature", y="Frequency", kind="scatter", ylim=[0,1])
plt.grid(True)
```



At first glance, the dependence does not look very important, but let's try to estimate the impact

of temperature t on the probability of O-ring malfunction.

1.2 Estimation of the temperature influence

Suppose that each of the six O-rings is damaged with the same probability and independently of the others and that this probability depends only on the temperature. If $p(t)$ is this probability, the number D of malfunctioning O-rings during a flight at temperature t follows a binomial law with parameters $n = 6$ and $p = p(t)$. To link $p(t)$ to t , we will therefore perform a logistic regression.

```
[24]: import statsmodels.api as sm

# We don't have 23 samples but 6 x 23

X = data[["Intercept", "Temperature"]]

y = data[["Malfunction", "Success"]]

logmodel = sm.GLM(
    y, X, family=sm.families.Binomial(sm.families.links.logit)
).fit()

logmodel.summary()
```

```
[24]: <class 'statsmodels.iolib.summary.Summary'>
      """
                                Generalized Linear Model Regression Results
=====
=====
Dep. Variable:      ['Malfunction', 'Success']    No. Observations:
23
Model:                                GLM    Df Residuals:
21
Model Family:                        Binomial    Df Model:
1
Link Function:                        logit    Scale:
1.0000
Method:                                IRLS    Log-Likelihood:
-15.823
Date:                                Thu, 08 Oct 2026    Deviance:
18.086
Time:                                10:36:31    Pearson chi2:
30.0
No. Iterations:                        6    Covariance Type:
nonrobust
=====
=====
                                coef      std err          z      P>|z|      [0.025      0.975]
-----
-----
```

Intercept	5.0850	3.052	1.666	0.096	-0.898	11.068
Temperature	-0.1156	0.047	-2.458	0.014	-0.208	-0.023

=====

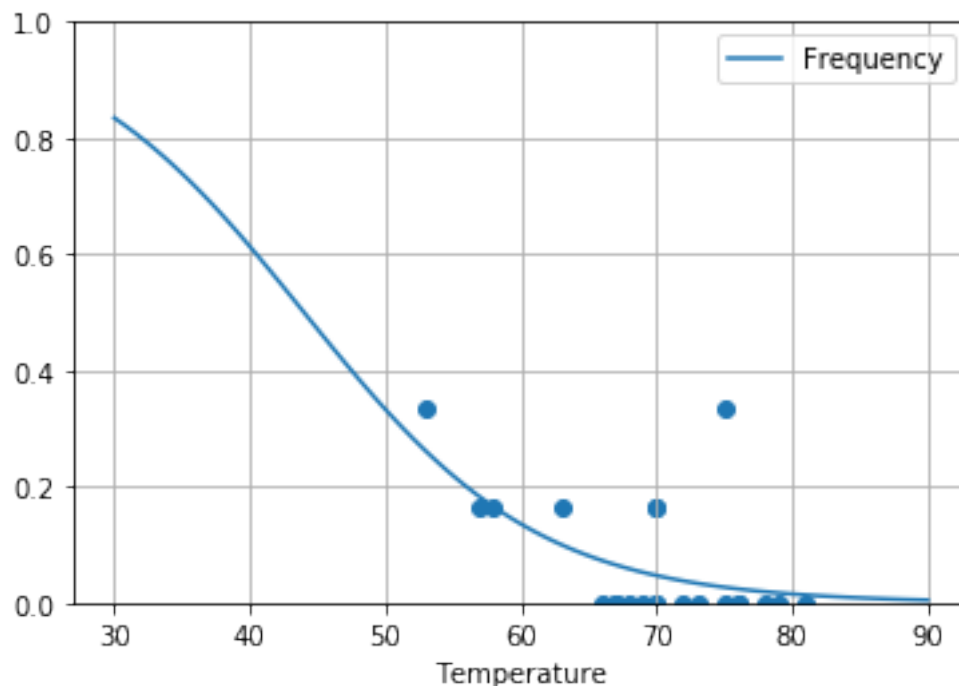
"""

We now have a temperature coefficient of -0.1156 and a p-value of 0.014, which means that we can consider that there's a significant negative link between temperature and O-ring malfunction (Lower temperature implies a higher probability of failure).

1.3 Estimation of the probability of O-ring malfunction

The expected temperature on the take-off day is 31°F. Let's try to estimate the probability of O-ring malfunction at this temperature from the model we just built:

```
[25]: %matplotlib inline
data_pred = pd.DataFrame({'Temperature': np.linspace(start=30, stop=90, num=121), 'Intercept': 1})
data_pred['Frequency'] = logmodel.predict(data_pred[['Intercept', 'Temperature']])
data_pred.plot(x="Temperature", y="Frequency", kind="line", ylim=[0,1])
plt.scatter(x=data["Temperature"], y=data["Frequency"])
plt.grid(True)
```



```
[26]: data = pd.read_csv("shuttle.csv")
print(np.sum(data.Malfunction)/np.sum(data.Count))
#We should change this model but idk how to do it
```

0.06521739130434782

This probability is thus about $p = 0.065$. Knowing that there is a primary and a secondary O-ring on each of the three parts of the launcher, the probability of failure of both joints of a launcher is $p^2 \approx 0.00425$. The probability of failure of any one of the launchers is $1 - (1 - p^2)^3 \approx 1.2$. That would really be bad luck.... Everything is under control, so the takeoff can happen tomorrow as planned.

But the next day, the Challenger shuttle exploded and took away with her the seven crew members. The public was shocked and in the subsequent investigation, the reliability of the O-rings was questioned. Beyond the internal communication problems of NASA, which have a lot to do with this fiasco, the previous analysis includes (at least) a small problem.... Can you find it? You are free to modify this analysis and to look at this dataset from all angles in order to to explain what's wrong.

```
[ ]: There is no reason in the part one to not take into consideration the
    ↳non-accidental data
because we want to study the link between, temperature and pressure
thus if we do this we only have a sample size of 7 which explain why the anova
    ↳test don't give any significative p-value.
After redoing the test we saw that there is a link between temperature and
    ↳crash.
```